



Hearing deficits of transformer-based multilingual ASR for anechoic and spatial signals

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Abstract

We compare automatic speech recognition (ASR) with human speech recognition (HSR) based on speech material that is traditionally used for diagnosing hearing deficits. Specifically, we quantify the human-machine gap with sentences in noise for different model sizes and two languages supported by Whisper, e.g., German and English. For German speech, we also determine the gap in different rooms and the presence of reverberation and localized noise. Results are put in context of audiological diagnosis using the speech reception threshold (SRT) (i.e., the SNR with 50% word recognition rate). We find that the largest ASR system is mildly hearing impaired when exposed to non-spatial, unpredictable US-English sentences and that using German speech degrades the SRT by 4.9 dB. In reverberant rooms, the gap reaches at least 5.9 dB. Based on the language bias, we estimate that same model can achieve better performance than normal-hearing listeners in anechoic conditions for US-English.

Index Terms: speech recognition, human-computer interaction, computational paralinguistics

1. Introduction

Normally hearing (NH) human listeners excel at recognizing speech and are able to communicate in complex acoustic conditions such as cocktail parties. In the past, this observation had motivated comparisons of human speech recognition (HSR) and automatic speech recognition (ASR) such as the classic review by [1] who reported that "Error rates of machines are often more than a magnitude greater than those of human for quiet, wide-band, read speech".

Comparisons between humans and machines were motivated by bridging the gap between two fields of research [2, 3], by improving ASR by learning from the biological counterpart [4, 5] and by simply measuring the gap to quantify how far we have come in closing the gap between HSR and ASR. Comparisons were conducted on sub-lexical level [4, 5, 6], based on word recognition or on sentence level. The above-mentioned research was conducted with ASR based on traditional architectures, i.e., a combination of Gaussian mixture models (GMMs) and hidden Markov models (HMMs).

In recent work, properties of ASR based on transformer-based E2E ASR has been analyzed, specifically, the open-source ASR engine Whisper by openAI which is capable of multi-lingual ASR [7]. A systematic analysis with different speech databases has shown that error rates obtained with Whisper strongly depend on accent, dialect, and gender (for instance, with error rates up to three times higher for accented speech compared to non-accented US English) [8]. Whisper and cloud-based ASR were benchmarked with *QuickSIN*, a simple speech-

in-noise test, indicating that Whisper *Base* could be considered as mildly hearing impaired, while larger Whisper models achieve the same or almost the same speech reception threshold (SRT), i.e., the signal-to-noise ratio (SNR) at which 50% of words are correctly identified [9]. This latter comparison was based on single-channel input without reverberation for accent-free US English. However, human listeners profit from spatial release from masking; their speech recognition performance increases when target speech and noise source are not co-located [10], which is an important aspect in the above-mentioned complex acoustic scenes. In an earlier human-machine comparison in spatial scenes, the SRT of an ASR system was more than 16 dB higher (worse) compared to human listeners [11], indicating that ASR could not profit from a spatial release of masking. That study was, however, based on a GMM-HMM system, and it is unclear how the current state-of-the-art compares to normal-hearing listeners.

In this paper, we measure the speech-in-noise recognition of Whisper ASR using an established audiological test, i.e., a matrix test that is based on noisy sentences. This test is available in several languages, which enables an analysis of the influence of input language to the ASR, in our case, US-English and German. To explore spatial release from masking in ASR, we also explore error rates in spatial scenes with a speaker at 0° azimuth and a localized masker that is positioned at different azimuth angles. These analyses are performed with different ASR models to study the effect of model size and its interaction with input languages and spatial speech signals.

2. Methods

2.1. Speech-in-noise tests

A widely used method for measuring the SRT relies on matrix tests with sentence lists specifically designed to prevent learning effects. The concept was first introduced by Hagerman [12] for Swedish, where sentences were generated using a random-walk process through a predefined word matrix. Wagener et al. [13, 14, 15] later refined this approach by developing a structured framework that ensured smooth word transitions, ultimately leading to the creation of the Oldenburg Sentence Test (OLSA).

These matrix tests consists of sentences that are syntactically identical (*<name><verb><number><adjective><object>*; e.g. 'Peter kauft drei alte Ringe'; 'Peter buys three old rings'), but are semantically unpredictable.

In listening experiments, these signals are mixed with a stationary noise with the same long-term spectrum as the test utterances [15]. During the HSR testing procedure, the SNR is modified according to the accuracy of word recognition in a noisy environment, aiming for a 50% correct response rate. This tech-

nique has been applied to develop similar tests in 20 languages, such as US-English [16], Dutch [17], Italian [18], and Spanish [19]. Further details on these tests can be found in [16]. The datasets used in this study are based on the US-English and German matrix tests.

2.2. Monaural anechoic data

The data for the monaural anechoic task used in this paper was first described in [15]. For this paper the speech was mixed with noise at 401 different SNRs in the range of -20 dB SNR up to 20 dB SNR in steps of 0.1 dB. For every SNR, ten sentences from one test list of the matrix sentence test are used. In total 4010 files with speech mixed with noise are generated. With this data, the psychometric function can be sampled. Later the SRT of the models should be compared to the NH listeners SRT. Measurements to obtain the NH SRT were conducted in [15].

The second dataset is similar to the one described above. It consists of 90 American-English sentences following the same syntax as the German sentences. A stationary noise following the long term spectrum of the American-English speech material is used. Ten sentences from the corpus were selected randomly and mixed at each analyzed SNR.

2.3. Binaural spatial data

To compare spatial release from masking and performance in different reverberant scenes, we used an existing dataset introduced by Beutelmann and Brand [20]. It features three spatial scenes (*Anechoic*, *Cafeteria*, *Office*) which are generated by convolving the monaural stationary noise and speech from the German matrix test with head related transfer functions (HRTF) recorded for different azimuth angles. HRTFs for *Office* and *Cafeteria* were measured by Beutelmann et al. [20], the anechoic HRTFs by Algazi et al. [21]. The latter are part of an open-source database and have been recorded with a KEMAR manikin. *Office* and *cafeteria* HRTFs were measured with a B&K 4128 manikin and using maximum length sequences. The loudspeakers in the office room had a distance of 1.45 m to the head center of the manikin, while in the cafeteria a single loudspeaker was positioned at various locations around the manikin seated at a table. In all scenes, the speaker is virtually placed at 0° azimuth, while the noise positions differ slightly due to practical considerations when measuring the HRTFs. (noise azimuth for *Anechoic* and *Office*: -140, -100, -45 0, 45, 80, 125, and 180°; noise azimuth for *Cafeteria*: -135, -90, -45, 0, 45, 90, 135, and 180°). To obtain the NH SRTs for the room conditions and angles, Beutelmann et al. [20] conducted measurements with eight NH participants. Speech intelligibility measurements were conducted using the HörTech Oldenburg Measurement Applications (OMA). One random test list containing 20 sentences was chosen to determine the SRT. Participants repeated the words they recognized, and responses were scored by an instructor. The speech level was adaptively adjusted to determine the SRT. A stationary noise was used as masker (cf. Section 2.1) and presented at a fixed level of 65 dB. Measurements were randomized and conducted in a laboratory setting.

2.4. ASR models

The models used in this paper are all Whisper [7] models of different size. The following model variables were examined: Tiny, Base, Small, large-v3-turbo (from now on referred to as

V3turbo). Due to computational constraints, the largest model available was not included. A medium-sized model was also not analyzed since pilot experiments indicated worse performance but higher computational cost than V3turbo. The implementation is part of the open-source framework Speechbrain [22] [23]. No changes were made to the architecture, nor was any training or adaptation carried out.

2.5. Determination of the speech reception threshold

For all acoustic conditions (including the monaural case and spatial scenes), audio is mixed with 121 different SNRs in the range of -40 to 20 dB SNR with steps of 0.5 dB SNR. 10 random audio files are selected and then mixed with noise at each SNR.

For spatial scenes only, the speech mixed with noise was then converted through a binaural speech intelligibility model (BSIM) from a stereo signal to a single channel signal as described below. This resulted in 9,680 signals for each spatial condition (29,040 in total).

We determined the SRT by calculating the word error rate (WER) for ten random sentences at different SNRs and by fitting a psychometric function (cf. Eq. 1) to the data and then obtaining the SNR with 50% error rate from the fit (with the free fit parameter a-d):

$$\Psi(SNR) = \frac{a}{1 + e^{4b(c-SNR)}} + d \quad (1)$$

Binaural integration: To obtain the single-channel data required as input to the ASR models, we integrated the input signals from the left and right ear in spatial scenes. To this end, we applied a blind equalization-cancellation (EC) processing stage borrowed from a model that predicts human speech recognition performance in spatial speech and noise conditions [24]. This EC stage can be regarded as a delay-and-subtract beamformer optimized for yielding human performance. This EC processing works blindly, that is, without any knowledge about the target speech, the interfering sounds, the room acoustics, or directions of sound sources. The internal parameters of the EC stage (i.e., the interaural gain and delay values that are applied before the cancellation stage) are controlled by a modulation analysis that maximizes the speech-similarity of the modulations of the single (mono) output signal. Consequently, this approach does only work for interfering sounds that are not modulated like human speech. In this study, only steady state noises were used.

3. Results

3.1. Monaural non-spatial signals

3.1.1. English speech signals

Table 1 summarizes the calculated SRTs and slopes of the anechoic condition for the English matrix sentence test. SRTs consistently improve with ASR model size. While the slope increases from Tiny to the Base model it does not change between the Base and Small. The biggest change can be seen between the slope of the Small and the V3turbo model. Figure 1 shows the fitted psychometric function for all models tested on English and German data. For comparison the NH psychometric curve is also shown for both languages. All the models reach a maximum recognition rate of 90% and more for the English input signals as shown in figure 1.

Table 1: *SRT and slope values of the English matrix sentence test for four different Whisper model sizes*

Model	SRT/dB	slope /(%/dB)
English		
Tiny	0.12	0.12
Base	-1.5	0.14
Small	-3.8	0.14
V3turbo	-5.2	0.21
German		
Tiny	6.5	0.098
Base	3.3	0.13
Small	0.6	0.17
V3turbo	-1.2	0.2

3.1.2. German speech signals

Table 1 shows the calculated SRTs and slopes for the German matrix test of the four models. Again, the SRT improves consistently with model size, as does the slope. The recognition performance saturates at different recognition rates depending on model size: V3turbo recognizes up to 90% of the words on average, while the smallest model saturates at 70%.

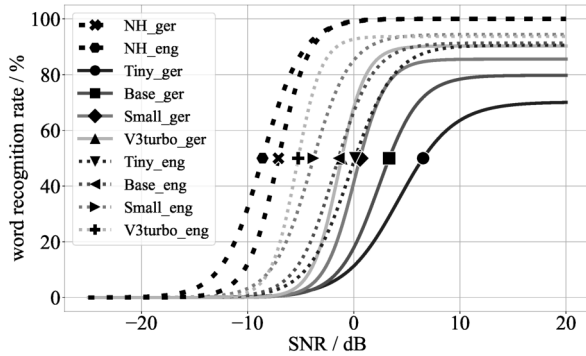


Figure 1: *Psychometric functions of normal-hearing listeners (dashed lines) and ASR systems with different model sizes. Data was obtained with dry, non-spatial sentences produced in English (dotted lines) or German (solid lines); markers denote the SRT.*

3.2. Binaural spatial signals

Figure 2 presents three subplots - one for each spatial condition for both HSR and ASR and all obtained with German speech data. SRTs for the listeners and all ASR systems peak at 0° azimuth (when speech and noise are co-located) and drops when the noise is moved to a different position. Hence, both HSR and ASR profit from spatial release from masking. The asymmetric shape of the data in *Cafeteria* arises from an acoustic asymmetrical setup with sound reflections from a window close to the recording microphone.

Results also show that larger ASR models consistently outperform the smaller ones. In the anechoic condition, V3turbo outperforms NH listeners for three azimuth positions of the noise. However, the performance of all ASR systems is heavily degraded in reverberant conditions: The best-performing ASR

Table 2: *Root mean squared errors (RMSE) and Pearson correlation values of the SRT predictions from multiple models for three different room conditions. The p-values associated with the correlation denote the significance level.*

		Room Conditions		
		Anechoic	Office	Cafeteria
Tiny	RMSE	15.6 dB	19.9 dB	17.4 dB
	r	0.88**	0.84**	0.94***
Base	RMSE	10.2 dB	15.4 dB	13.5 dB
	r	0.84**	0.86**	0.94***
Small	RMSE	4.7 dB	11.1 dB	10.3 dB
	r	0.89**	0.95***	0.93***
V3turbo	RMSE	2.8 dB	8.6 dB	8.0 dB
	r	0.87**	0.95***	0.95***

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

system still produces an average HSR-ASR gap of 7.9 dB and 8.6 dB for *Cafeteria* and *Office*, respectively.

Table 2 shows the RMSE and Pearson correlation between every model and the normal-hearing listeners. The correlation is invariant to shifts of the HSR and ASR curves, but is used to quantify the relative improvement by spatial release from masking. Correlation values are relatively stable across ASR models, while the RMSE consistently improves with model size. Correlation values are also consistently *high*, which means that ASR profits from masking release similarly to human listeners.

Estimating ASR performance for US-English: Listeners' data in spatial scenes was only available in German for this study. Since we determined the bias between US-English and German in monaural conditions, we can estimate the ASR performance in spatial scenes by shifting the ASR data points from Figure 2 by the model-specific bias for the two languages (Table 2). At the same time, we lower the listeners' SRT by 1.5 (the difference for NH listeners for the English and German matrix test). With these corrections, Whisper V3turbo is better by 1.9 dB on average than HSR in *Anechoic*. In reverberant conditions, it is still 6.1 and 5.4 dB worse on average (for *Office* and *Cafeteria*).

4. Discussion

4.1. Monaural non-spatial signals

The SRT of the US-English matrix test with a female speaker in an anechoic condition is -8.6 dB for NH listeners, which is lower (better) than for all ASR models explored here. The best-performing ASR system produces an SRT which is 3.4 dB higher than the reference for NH listeners. A study that analyzed the SRT and the related hearing loss reported an SRT increase by 0.1 dB for 1 dB of elevated hearing threshold [25]. From this value, we can roughly estimate that the performance gap between HSR and ASR corresponds to a hearing loss of $3.4/0.1 \text{ dB} = 34 \text{ dB}$, i.e., a mild hearing loss according to HL profiles suggested by Bisgaard [26]. We do not want to imply that E2E ASR suffers from sensorineural HL, which is often the cause of elevated hearing thresholds. We assume that factors that relate to the implicit language model play a role, as discussed below.

All Whisper models perform better when using English input data compared to German. This is expected as they were

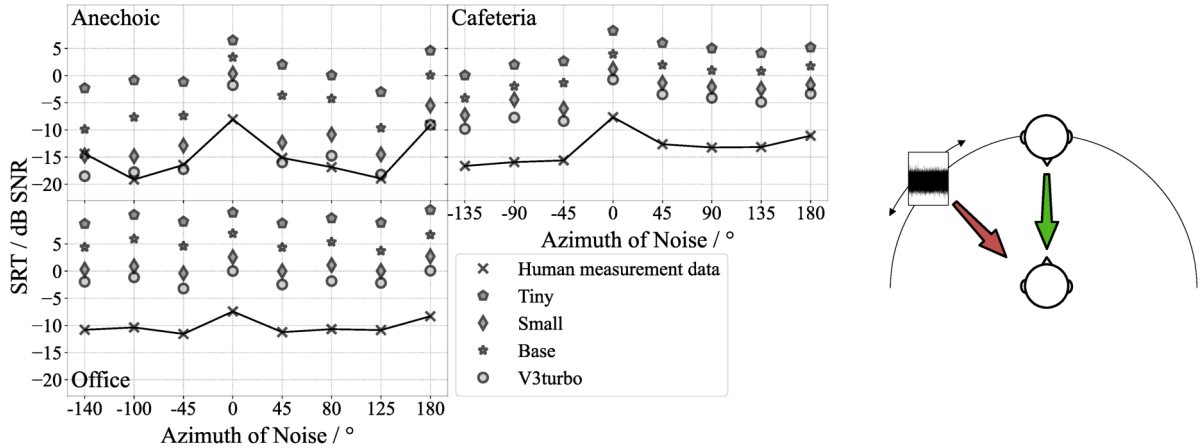


Figure 2: SRTs of human listeners and ASR models of different complexity in three rooms. Noisy sentences are presented from 0° azimuth while the localized noise has varying positions as illustrated.

trained on more English data than on data from any other language (680 thousand hours of training data had 117 thousand hours of multilingual training data included) [7]. Still, all ASR models fail to reach NH HSR performance. Compared to the German input data, the difference between the Small model’s SRT and the NH SRT (-4.8 dB) is smaller than the difference between the V3turbo model’s SRT and NH SRT for German input speech (-5.8 dB). Hence, even the smaller model performs better on monaural anechoic English data than the bigger model on the equivalent German data. When looking at the calculated slopes of the psychometric fits for the models it can be seen that for the German data the Small and V3turbo model achieve equally high or higher slopes than the psychometric function of NH listeners. Typically, the slopes are clearly decreasing with increasing (worse) SRT in real-world measurements for listeners with a hearing-impairment. Something similar can be observed for ASR, since the better-performing systems also result in steeper slopes. However, the best models show even steeper slopes than the NH listeners, despite the better SRT of the NH listeners. These high slopes indicate that the model recognition performance is increasing rapidly over a smaller SNR range. Both matrix tests applied in this study have been optimized for steep slopes in NH listeners by maximizing the homogeneity of the intelligibility of all test words. The steep slopes in the ASR systems indicate that this homogeneity is also given from the ASR perspective, which should be interesting when using ASR to optimize or analyze speech material that might be used later in listening experiments.

4.2. Binaural spatial signals

In condition *Anechoic*, Whisper V3turbo is on par (and in some conditions better) than HSR. We also estimated that super-human performance can be obtained for English speech signals. However, reverberant conditions have a strong detrimental effect on the ASR performance, reflected in the large shift between ASR and HSR in Figure 2. When using Whisper in these conditions, algorithms for dereverberation could be applied [27, 28]. Alternatively, ASR models could be exposed to more spatial data during training, which would presumably lower the corresponding SRTs.

The spatial benefit is similar for HSR and ASR, as indicated

by the parallel shape of predictions and measurements in Figure 2 as well as by the high correlation coefficients in Table 2. This suggests that ASR takes advantage of spatial release from masking in a similar fashion as HSR, alas on a different overall performance level.

4.3. The effect of unpredictable sentences

In a related study [9], it was shown that Whisper is normal-hearing according to the QuickSIN test (non-spatial, US-English), except for the smallest model (Whisper Base) analyzed in the paper. This is in contrast to our finding with SRT differences of at least 3 dB. We assume that the language model implicitly learned by the ASR system causes this difference: QuickSIN contains sentences for which individual words can be predicted from the context (such as “The lake sparkled in the red hot sun” or “Fake stones shine but cost little”). For matrix tests, the 10 alternatives for each word position are equally likely, resulting in a much higher perplexity than for QuickSIN.

In other words, HSR outperforms ASR for unpredictable words, but HSR-ASR gap vanishes when moving to predictable words (QuickSIN). This implies that ASR has a stronger benefit from the language model than human listeners when switching from matrix sentences to QuickSIN. This suggests that Whisper models could profit from improvements that relate to signal, word or feature level (which could be as easy as exposing the model to additional noise during training).

5. Summary

This paper compared HSR with multilingual Whisper ASR using transformer architectures based on a speech-in-noise test with semantically unpredictable sentences. For this speech material, we found that the largest Whisper model with English input signals produces recognition scores with slightly elevated SRTs (which are equivalent to a mild hearing loss). When using German speech as input, SRTs degrade by 4.9 dB compared to US English; we found the smallest ASR system to perform better with English than the second largest model for German. All ASR models profit from spatial release from masking, and ASR performs better than humans in some anechoic conditions with spatial noise. However, SRTs are higher by at least 5.9 dB for ASR model in reverberant conditions.

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